

Panama Sum (panama)

While their car was lost in Panama, Carlos and Pablo decided to have fun with some math.



Figure 1: The city of Panama.

One day, Carlos came up with what he called a *Panama sum*. Basically, a *Panama sum* is the sum of some values in an 1-indexed array, where the values on the odd positions are added to the sum and the values on the even positions are subtracted from the sum. For example, the *Panama sum* of the array $[10, 6, 9, 4, 2, 0]$ is $10 - 6 + 9 - 4 + 2 - 0$ which is equal to 11.

Because finding this is way too easy for Pablo (he is a programmer), he decided to make this challenge more difficult. Therefore, he invented the *maximum Panama sum*, which is defined as the *maximum Panama sum* we can get if we select a non-empty contiguous subarray from a given array.

Now he wants Carlos to find this value for multiple queries which might involve changing some values as well. Help Carlos get this problem done and he will take you for a ride once he finds his car!

 Among the attachments of this task you may find a template file `panama.*` with a sample incomplete implementation.

Input

The first line contains the integers N and Q . The second line contains N integers V_i .

Then Q lines follow, each containing a query. A query can be one of the following:

- 1 a b : Change the value of V_a to b .
- 2 l r : Find the *maximum Panama sum* of the subarray $V_l, V_{l+1}, \dots, V_r$.

Output

You need to write k lines, where k is the number of queries of type 2. Each line should contain a single integer, the answer to the i -th query.

Constraints

- $1 \leq N \leq 100\,000$.
- $1 \leq Q \leq 100\,000$.
- $0 \leq V_i \leq 10^9$ for $i = 1 \dots N$.
- $1 \leq a \leq N$ and $0 \leq b \leq 10^9$ for each query of type 1.
- $1 \leq l \leq r \leq N$ for each query of type 2.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (25 points) $N \leq 100$ and $Q \leq 1000$.

- **Subtask 3** (35 points) $N \leq 5000$ and $Q \leq 5000$.

- **Subtask 4** (40 points) No additional limitations.


Examples

input	output
10 6 5 9 1 2 3 4 6 4 2 8 2 3 7 2 4 10 2 6 9 1 5 8 2 1 10 2 4 6	6 10 6 10 8